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# generate_dataset.py
#
# I could not find a clean, ready-to-use
spam dataset that I was allowed
# to redistribute for this project, so I
decided to build my own small
# dataset instead. It is made by combining
a few sentence templates with
# random words/numbers, so it is not real
email data, but it gave me
# something to actually train and test the
model on.
#
# I know a real dataset (like the SMS Spam
Collection dataset) would be
# more realistic, and I mention that in my
README as a future improvement.

import csv
import random

random.seed(1) # keeping this fixed so I
get the same dataset every time I run it

spam_start_words = [
    "CONGRATULATIONS", "URGENT", "WINNER",
    "FINAL NOTICE",
    "Limited Time Offer", "ATTENTION",
    "FREE GIFT", "You have been selected"
]
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spam_middle_parts = [  
    "you have won a ${amount} gift card,  
    click here to claim it",  
    "your account will be suspended unless  
    you verify your details",  
    "claim your free {item} today, no  
    purchase needed",  
    "you have been selected for a cash  
    reward of ${amount}",  
    "get a free {item}, limited stock,  
    order now",  
    "your package could not be delivered,  
    click to reschedule",  
    "earn ${amount} a day from home, no  
    experience needed",  
    "your loan of ${amount} has already  
    been approved",  
    "we noticed unusual activity, confirm  
    your password here",  
    "this is your last chance to claim  
    ${amount}",  
]  
  
spam_items = ["iphone", "laptop", "gift  
card", "vacation package", "smart watch"]  
  
spam_end_parts = [  
    "Click here to claim:  
    http://bit.ly/claim-now",  
    "Reply YES to claim your prize",  
    "Visit our website now",  
    "Call this number immediately",  
    "Offer ends soon, do not miss it",
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]

# these are just normal, everyday messages
a student or working person might send
ham_messages = [
    "Hey, are we still meeting for lunch
tomorrow?",
    "Can you send me the notes from today's
class?",
    "Don't forget we have the project
submission on Friday.",
    "Thanks for helping me with the
assignment yesterday.",
    "What time does the lab open
tomorrow?",
    "I finished the report, will send it
tonight.",
    "Can we reschedule our call to 5pm?",
    "Happy birthday! Hope you have a great
day.",
    "The meeting got moved to next
Monday.",
    "Let me know if you're free this
weekend.",
    "I'm stuck on question 3, can you help
me out?",
    "Just checking in, how did the
interview go?",
    "The wifi in the library is down
again.",
    "See you at the group study session at
6.",
    "Thanks for the ride home yesterday.",
    "Can you review my code before I submit
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it?",
    "My train got delayed, I'll be 15
minutes late.",
    "Reminder: assignment 2 is due at
midnight.",
    "Nice presentation today, learned a
lot.",
    "Let's grab coffee sometime this
week.",
]

def make_spam_message():
    start = random.choice(spam_start_words)
    middle =
random.choice(spam_middle_parts).format(
    amount=random.choice([100, 250,
500, 1000]),
    item=random.choice(spam_items)
)
    end = random.choice(spam_end_parts)
    return f"{start}: {middle}. {end}"

def make_ham_message():
    # just picking one of the normal
messages, sometimes I add a name at the
front
    message = random.choice(ham_messages)
    if random.random() < 0.3:
        name = random.choice(["Alex",
"Sam", "Priya", "John", "Maria"])
        message = f"Hi {name}, " +
message[0].lower() + message[1:]

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        return message

def main():
    rows = []

    # 200 of each felt like enough to train
    something without the file getting huge
    for _ in range(200):
        rows.append(("spam",
make_spam_message()))
    for _ in range(200):
        rows.append(("ham",
make_ham_message()))

    random.shuffle(rows)

    with open("spam_dataset.csv", "w",
newline="", encoding="utf-8") as f:
        writer = csv.writer(f)
        writer.writerow(["label", "text"])
        writer.writerows(rows)

    print(f"Created spam_dataset.csv with
{len(rows)} messages")

if __name__ == "__main__":
    main()
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